SOP-COMPUTER SYSTEM VALIDATION

[CONFIDENTIAL]
[UTTHUNGA]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
01	Issue B Version 0.1	Draft	Implementation of the Integrated Management System	Sachin Aramani 27-Aug-24		
02	Issue B, Version 1.0	New Process Implementation			Sanmangala M S 09-Sep-24	Sanmangala M S 11-Sep-24
03	Issue B Version 1.1	Document formatting done as per document guidelines	Updated in line with the document guidelines	Obul Prasad Reddy 20-Dec-24	Anitha N 23-Dec-2024	Santosh V 31-Dec-2024
04	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-Aug-2025	Santosh V 08-Aug-25	Sanmangala MS 14-Aug-2025

1 Purpose

The purpose of this SOP is to establish a standardized process for Computer System Validation (CSV) within Utthunga to ensure that all computer systems used in regulated activities are validated in compliance with relevant regulatory requirements. This SOP ensures that these systems operate as intended, maintain data integrity, and meet the requirements of 21 CFR Part 11.

2 Scope

This SOP applies to all computer systems used within Utthunga that are involved in the creation, modification, maintenance, archiving, retrieval, or transmission of electronic records and electronic signatures. This includes systems used in laboratory, manufacturing, quality management, clinical trials, and other regulated activities.

3 Abbreviations

a. CSV: Computer System Validation

b. SOP: Standard Operating Procedure

c. IQ: Installation Qualification

d. OQ: Operational Qualification

e. PQ: Performance Qualification

f. SDLC: Software Development Life Cycle

g. GxP: Good Practice guidelines and regulations

h. IT: Infrastructure Maintenance

i. PM: Project Manager

j. DM: Delivery Manager

k. BU Head: Business Unit Head

4 Terms and Definitions

a. Computer System - Any automated system that collects, stores, processes, or transmits data.

b. CSV: Computer System Validation – A documented process of ensuring that a computer system meets its intended use and complies with regulatory requirements.

c. IQ: Installation Qualification – Verification that the system is installed correctly according to the specifications.

d. OQ: Operational Qualification – Verification that the system operates as intended under all specified operating conditions.

e. PQ: Performance Qualification – Verification that the system performs effectively and reproducibly under real-world conditions.

f. GxP: Good Practice – A general term for regulations and guidelines in areas such as manufacturing (GMP), clinical trials (GCP), and laboratories (GLP).

5 Responsibilities

a. Project Team: Responsible for planning, executing, and documenting the validation activities.

b. System Owner: Ensures the system meets user requirements and is maintained in a validated state.

c. Project Manager: Prepare requirement, verification, validation documentation and ensures compliance with regulatory requirements.

d. IT Department: Provides technical support and ensures system infrastructure is maintained

e. Delivery Manager: Reviews and Approves requirement, verification, validation documentation and ensures compliance with regulatory requirements.

f. BU Head: Reviews and Approves requirement, verification, validation documentation and ensures compliance with regulatory requirements.

6 RACI

Process	Responsible	Accountable	Consulted	Informed
Validation Planning	PM	DM	Customer	BU Head
Requirement Specification	Project Team	PM	Customer	BU Head, DM
Design and Development	Project Team	PM	BU Head, DM	Customer
Installation Qualification	Project Team	PM	BU Head, DM	Customer
Operational Qualification	Project Team	PM	BU Head, DM	Customer
Performance Qualification	Project Team	PM	BU Head, DM	Customer
Validation Report and System Release	Project Team	PM	BU Head, DM	Customer
Post-Validation Activities	Project Team, Customer	Project Manager	BU Head, DM	Quality

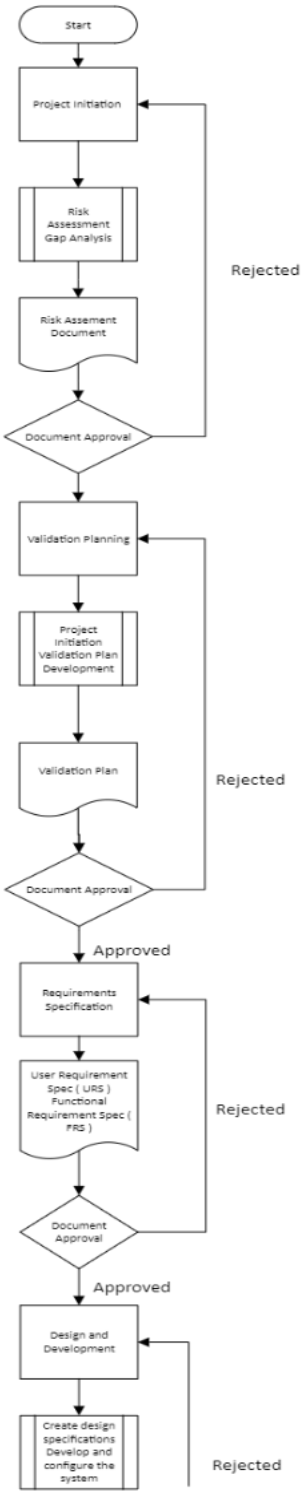
7 SIPOC

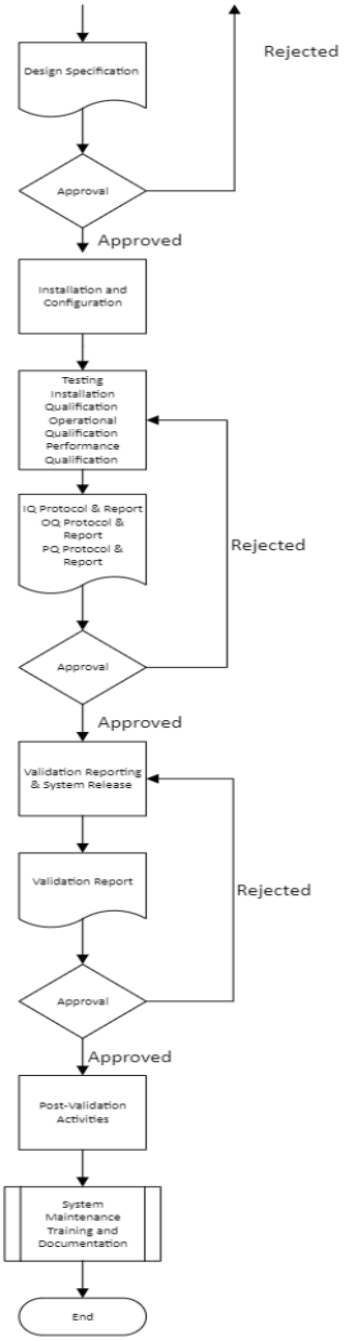
Supplier	Input	Process	Output	Customer
Regulatory Bodies (e.g., FDA, EMA)	Regulatory Guidelines (e.g., 21 CFR Part 11, GAMP 5)	Validation Planning	Validation Plan	Project Team
Clients	User Requirements	Requirement Specification	User Requirements Specification (URS)	Development Team

			Functional Requirements Specification (FRS)	Testing Team Team
Software Vendors	Software and System Specifications, URS, FRS	Design and Development	Design Specification (DS) Developed System	Testing Team
IT Department	Hardware and Infrastructure Components	Installation Qualification	Installed and Verified System IQ Test Results IQ Report	Testing Team
Development Team	Configured/Developed System	Operational Qualification (OQ)	OQ Test Results OQ Report	Testing Team
Testing Team	OQ Approved System	Performance Qualification (PQ)	PQ Test Results PQ Report	Customer
Testing Team	IQ, OQ, PQ Reports	Validation Reporting and System Release	Validation Summary Report Approved Validated System	Business Unit Head Customer
Testing Team	Validated System	Post Validation Activities	Maintenance Plan Training Materials	Customers

8 Overall Process Flow Chart

Computer System Validation




Design Specification
Rejected
Approval
Approved
Installation and Configuration
Testing
Installation Qualification
Operational Qualification
Performance Qualification
IQ Protocol & Report
OQ Protocol & Report
PQ Protocol & Report
Rejected
Approval
Approved
Validation Reporting & System Release
Validation Report
Rejected
Approval
Approved
Post-Validation Activities
System Maintenance Training and Documentation
End

9 Process Description

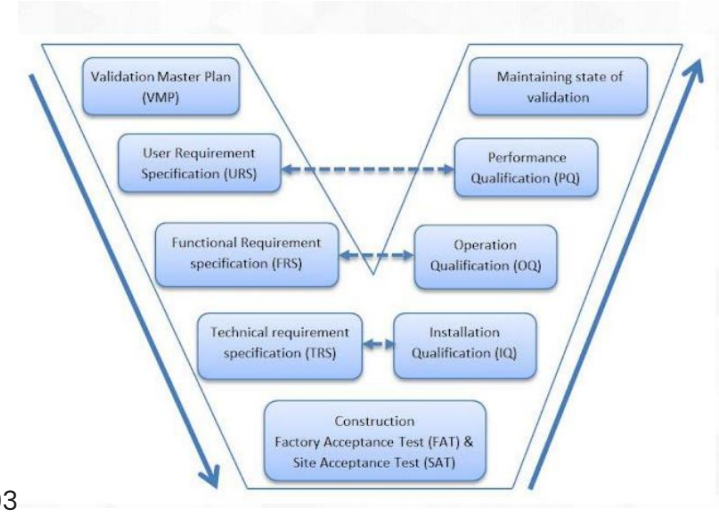
9.1 Project Initiation

a. Conduct risk assessment to identify the potential risks associated with the system and document the risks as per SOP - Risk & Opportunity Management.

b. Conduct Gap analysis to compare system requirements against regulatory requirements

9.2 Development Methodology

a. Development Methodology will be followed as per V Model Guidelines -



UTSDDMGDL03

9.3 Development

a. Validation Planning

- Develop a validation plan outlining the scope, objectives, responsibilities, timeline, and deliverables.
- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for Validation Plan document.

b. Requirements Specification

- Document the User Requirements Specification (URS)
- Document the Functional Requirements Specification (FRS) by detailing the specific functionalities that the system must perform.

- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for Requirement Specification document.

c. Design and Development

- Document the system architecture, components, and interfaces in design specification (DS)
- Implement version control and change management processes as per SOP - Configuration Management & SOP - Change Management.
- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for Design document.
- Develop the system as per requirements and conduct code review as per SOP - Review Process.

d. Installation and Configuration

- Ensure that the system meets hardware and software requirements, and back up any existing data if upgrading.
- Run the installer, configure basic settings (e.g., installation directory), and install any necessary dependencies.
- Set up user roles and control access levels.
- Configure software-specific settings like database connections, security, network, and backup options.
- Set access controls, logging, and monitoring to ensure system security and operational transparency.

9.4 Testing

a. Installation Qualification (IQ)

- Verify that the system is installed according to the manufacturer's specifications.
- Ensure the operating environment (e.g., hardware, software, network) is suitable and properly configured.

- Record all installation activities and outcomes in IQ Protocol and IQ Report.
- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for IQ Protocol and IQ Report document.

b. Operational Qualification (OQ)

- Test the system's functionalities by conducting functional Testing to ensure they operate as specified in the FRS.
- Conduct security testing to verify access controls, user roles, and audit trails.
- Record all Operational testing activities and outcomes in OQ Protocol and OQ Report.
- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for OQ Protocol and OQ Report document.

c. Performance Qualification (PQ)

- Validate that the system performs reliably under real-world conditions.
- Conduct stress testing to test the system's performance under extreme conditions to ensure stability.
- Record all performance testing activities and outcomes in PQ Protocol and PQ Report.
- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for PQ Protocol and PQ Report document.

d. Validation Reporting and System Release

- Compile a validation report summarizing all validation activities, results, and deviations.
- If any deviation from desired requirements, customer signoff has to be taken prior to the release.
- Obtain signatures from relevant stakeholders as per defined responsibilities in section 5 for Validation Report document.

- Release the system to relevant stakeholders.

e. Training

- Ensure users are trained on system operation and compliance requirements.
- Provide training materials to the users.

f. Monitoring and Review Activities

- Authorize the system for use, based on successful validation.
- Conduct monthly reviews well within AMC Period to ensure the system remains in a validated state.

10 Information Security requirements

a. Ensure that all systems and environments used during CSV are controlled and secured according to ISO 27001 standards.

b. Implement strict access control measures during CSV to ensure that only authorized personnel can access sensitive data and systems.

c. Establish a change management process for CSV, ensuring that all changes are evaluated for information security risks.

d. Ensure that system configurations are securely managed and documented throughout the validation process.

e. Develop and implement an incident management process for handling information security breaches during CSV.

11 References

a. SOP-Risk & Opportunity Management (RMP) – UTIMRMPSOP V1.2

b. SOP-Configuration Management UTPSCMPSOP01 V1.3

c. SOP-Change Management UTPSCMPSOP02 V1.3

d. SOP-Review Process (RVP)-UTPSRVPSOP V1.0

e. V Model Guidelines - UTSDDMGDL03 V1.0

f. 21 CFR Part 11: Electronic Records and Electronic Signatures (Amended version 09/06/24)

g. GAMP 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition)

12 Metrics

Metric	Formula
Validation Coverage	No of Validated system or Components / Total No of System or Components Required Val
Defect Detection Ratio	Number of Defects Identified / Total No of Test Cases Executed
Validation Cycle Time	End Date of Validation – Start Date of Validation

13 Annexures

NA

14 Forms

a. System Design Specification Template UTSDCSVTMP01

b. Validation Plan UTSDCSVTMP10

c. Validation Summary Report UTSDCSVTMP11

d. Installation Qualification Protocol UTSDCSVTMP02

e. Installation Qualification Report UTSDCSVTMP03

f. Operational Qualification Protocol UTSDCSVTMP04

g. Operational Qualification Report UTSDCSVTMP05

h. Performance Qualification Protocol UTSDCSVTMP06

i. Performance Qualification Report UTSDCSVTMP07

j. Factory Acceptance Test UTSDCSVTMP08

k. Site Acceptance Test UTSDCSVTMP09

l. Deviation Form –UTIMMSPTMP07